EXP 7: Write a program to implement Logistic Regression (LR) algorithm in python

CODING:

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import math

# Training Dataset
X = [
   [1, 2], [2, 3], [3, 2],
   [6, 7], [7, 8], [8, 7]
]

Y = [0, 0, 0, 1, 1, 1]

# Initialize Weights and Bias
weights = [0.0, 0.0]
bias = 0.0
learning_rate = 0.01

# Sigmoid Function
def sigmoid(z):
   return 1 / (1 + math.exp(-z))

# Training the Model
for epoch in range(5000):
   dw = [0.0, 0.0]
   db = 0.0

   for x, y in zip(X, Y):
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        z = sum(weights[i] * x[i] for i in range(2)) + bias
        prediction = sigmoid(z)

        error = prediction - y

        for i in range(2):
            dw[i] += error * x[i]

        db += error

    # Update Weights and Bias
    for i in range(2):
        weights[i] -= learning_rate * dw[i] / len(X)

    bias -= learning_rate * db / len(X)

# Display Model Parameters
print("Weights:", weights)
print("Bias:", bias)

# Prediction Function
def predict(sample):
    z = sum(weights[i] * sample[i] for i in range(2)) + bias
    probability = sigmoid(z)

    if probability >= 0.5:
        return 1
    else:
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    return 0


# Test the Model
test_data = [[2, 2], [7, 7], [3, 3], [8, 8]]

print("\nLogistic Regression Results:")

for sample in test_data:
    result = predict(sample)
    print("Input:", sample, "Predicted Class:", result)
```

OUTPUT:

